



Sea slug inspired smart marine antifouling coating with reversible chemical bonds: Controllable UV-responsive coumarin releasing and efficient UV-healing properties

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ABSTRACT

Poly(dimethylsiloxane) (PDMS) has been received widespread attention as marine antifouling coatings, while its controllable release performance of antifoulants and mechanical behavior are relatively insufficient. In this study, inspired by sea slug's defensive behavior against predators by ingesting toxins from prey and then releasing these toxins, a PDMS-based smart marine antifouling polyurethane (PU) coating was prepared, which had UV-responsive controllable coumarin-releasing and UV-healing properties. The UV-responsive coumarin-releasing mechanism was based on the dimerization and de-dimerization of reversible chemical bond between two coumarin groups. As a sort of eco-friendly antifoulant, modified coumarin and eugenol were fixed in PU chain by polycondensation. Then additional free coumarin was "locked" with the fixed coumarin under 365 nm UV and released stably in response to 254 nm UV. The coating could also efficiently self-heal under 365 nm UV through dimerization between fixed coumarin groups in polymer chains. Moreover, it had satisfactory mechanical behavior, transparency, antibacterial property, anti-algae property and nine-month marine field antifouling performance, which demonstrated the promising application in marine and other antifouling fields.

1. Introduction

Biofouling always has negative influence on marine industry, clinical medicine, biological engineering and other fields. Especially, marine fouling does great harm to surface of marine equipment, increasing sailing resistance and fuel consumption [1]. As the tributyltin (TBT) was forbidden being added in marine coatings because of its high teratogenicity, environmentally friendly marine antifouling coatings have been extensively studied for many years [2].

Many antifouling coatings are used in the battle against marine fouling, such as TBT free self-polishing coating [3], low surface energy (SE) fouling-release coating [4-6], micro-structured surface coating [7-11], hydrophilic coating [12-15], amphiphilic coating [13,16-19], antifoulant-contain coating [20-24], biodegradable coating [3,25,26], coating with hydrolysable main and branch chain [3,27-29], slippery

liquid-infused porous surface (SLIPS) [30-34], anti-bacterial coating [35-38], mimic peptide coating [39], fluorescent coating [40] and so on. Among these materials, polydimethylsiloxane (PDMS) has attracted continuous attention as antifouling coatings because of low SE, low Young's modulus and relatively economical cost [2]. However, PDMS is easy to be damaged and difficult to be repaired or healed [41], which restricts its application in complex marine environments [2]. In addition, the removal of fouling organisms on PDMS coating largely depends on the scouring and shearing effect of sea water, so it shows inefficient performance in static antifouling applications [42].

In order to overcome this disadvantage, the usual solution is to add antifouling agents, such as triclosan, DCOIT(4,5-dichloro-2-n-octyl-4-isothiazolin-3-one), Cu₂O, medetomidine and diketene. Antibacterial natural products, such as coumarins and eugenols, have the potential to be used as environmentally friendly antifoulants in marine coating

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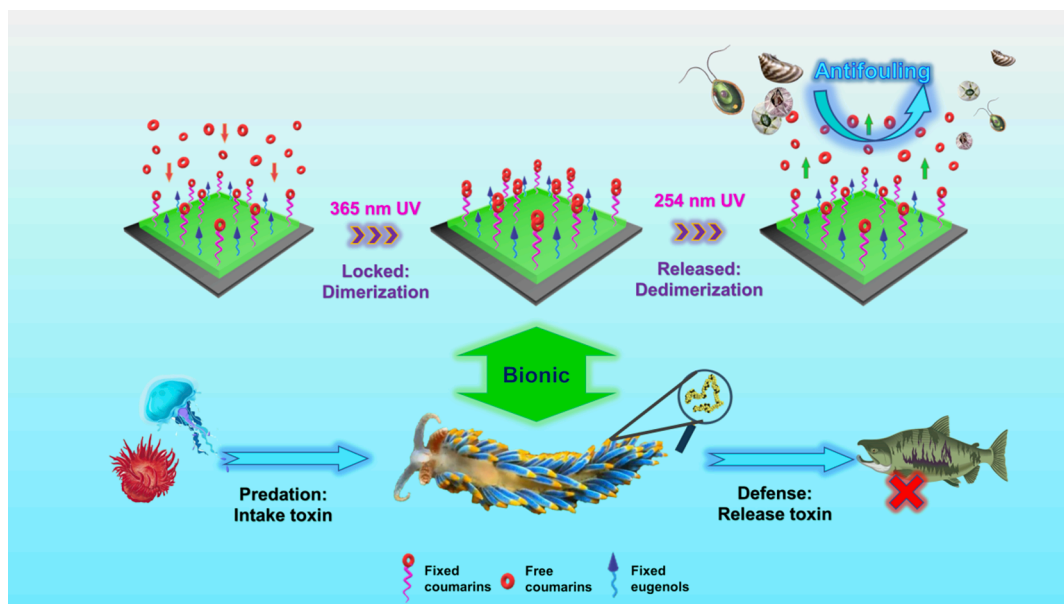
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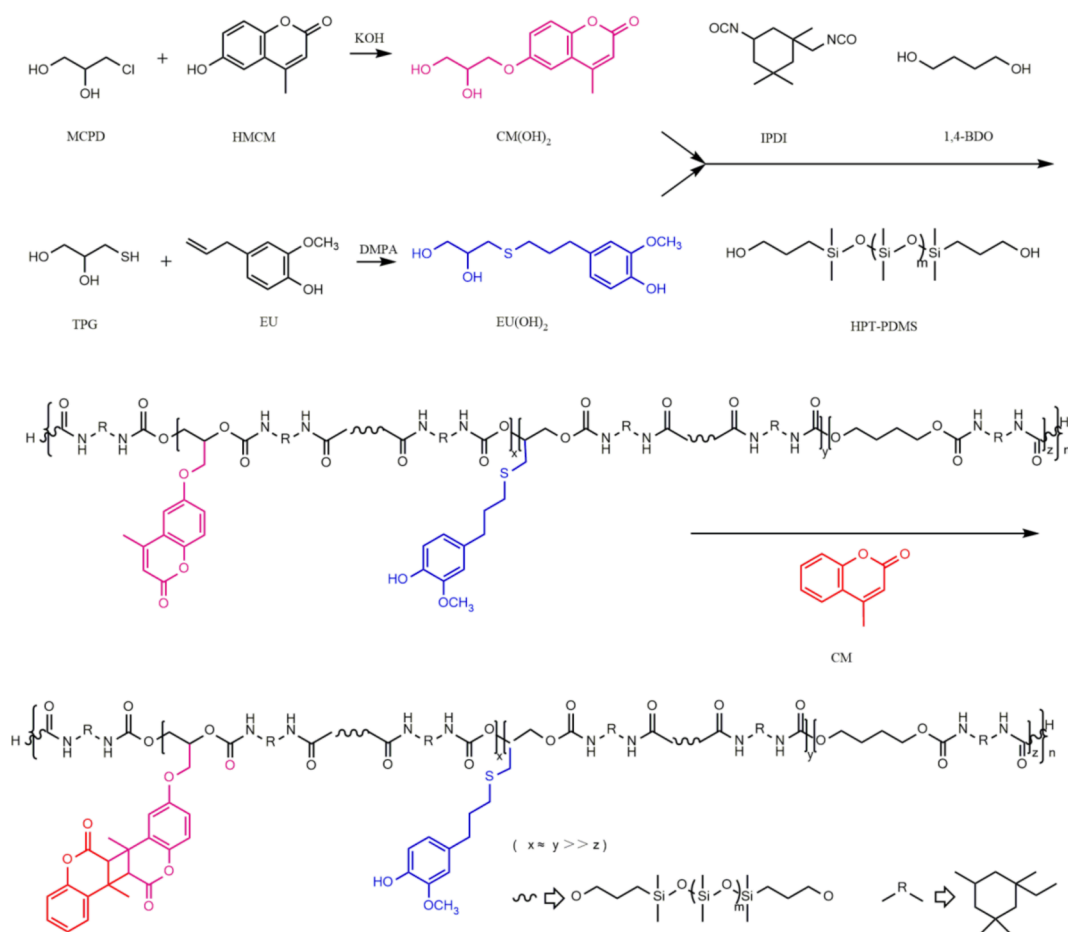
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Scheme 1. Design and mechanism of PDMS-based smart marine antifouling coating.



Scheme 2. Synthesis process of antifouling coating.

[23,52]. As the antifouling agent are physically mixed into the coating, the releasing rate is difficult to be controlled. The polymer capsule shell can be used to encapsulate the antifoulants, which hydrolyzed or ruptured by external stimuli to control the release of the antifouling

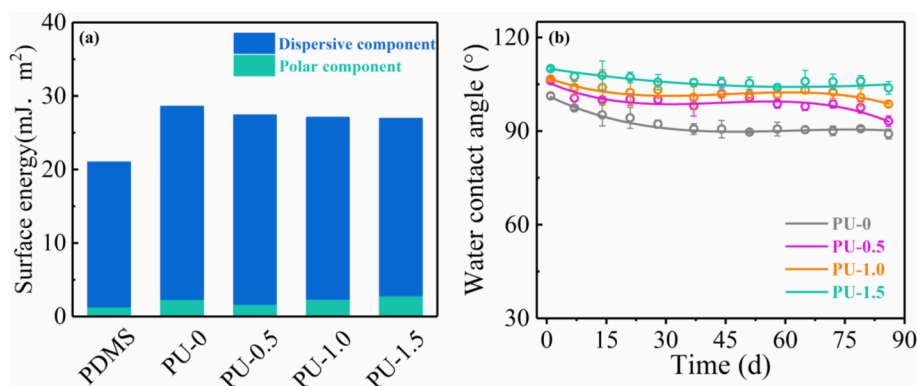
agent [1]. While addition of capsules will increase the surface energy and modulus, and the preparation of hollow capsules is relatively complicated and not easy for large-scale applications.

Another defect of PDMS is weak mechanic performance. To

Table 1

The feed ratio of samples.

Samples	HDT-PDMS /mmol	IPDI /mmol	CM(OH) ₂ /mmol	EU(OH) ₂ /mmol	CM /mmol	1,4-BDO /mmol	CM(OH) ₂ & EU(OH) ₂ /HDT-PDMS
PU-0.5	1	2	0.25	0.25	0.25	0.1	0.5
PU-1.0	1	2	0.5	0.5	0.5	0.1	1.0
PU-1.5	1	2	0.75	0.75	0.75	0.1	1.5
PU-0	1	2	0	0	0	1.1	0
PU-CM	1	2	0.5	0	0.5	0.51	0.5
PU-EU	1	2	0	0.5	0	0.51	0.5

**Fig. 1.** (a) SE of modified PU and (b) WCA trends in ASW during 3 months.

overcome this shortcoming, polyurethane (PU) [22], polyurea (PUa) [43], nanofillers (SiO₂, TiO₂, graphene, etc.) [44–47] or reversible bonds (hydrogen bonds, disulfide bonds, supramolecular forces, etc.) [43,48,49] are introduced into PDMS to enhance self-healing and mechanical properties. For example, dimerization mechanism has been applied to increase self-repair performance of PDMS [50,51].

In the present study, we have synthesized a bionic marine antifouling coating by imitating the defense behavior of sea slug, as the scheme 1 showed. Sea slug preys on poisonous sea anemones and jellyfish, absorbing and storing their toxins or stinging cells, transferring them to epidermis. After being stimulated by external mechanical and light from its predators, sea slug will activate stinging cells and release toxins to defend against them [53]. Similarly, a PDMS-based smart marine antifouling coating was prepared, which had UV-responsive controllable coumarin-releasing and UV-healing properties. The UV-responsive coumarin-releasing mechanism was based on the dimerization and de-dimerization of reversible chemical bond between two coumarin groups. Modified coumarin and eugenol were fixed in polymer chain to enhance the antifouling performance of coating body. Then additional free coumarin was “preyed” into the polymer “sea slug” by photo-dimerization under 365 nm UV and released from “sea slug” by stimulation of 254 nm UV to defend against “predators”. By imitating the behavior of sea slug’s defensive behavior, we provided a solution to control the release performance of antifouling agents. Moreover, the photo-dimerization of coumarin fixed on polymer chains endowed coating with UV-healing property. The PDMS-based coating prepared here has proper SE (20–30 mJ·m⁻²), low Young’s modulus (<4.5 MPa), tough adhesion (>1 Mpa), remarkable antibacterial and anti-algae property, showing satisfactory marine antifouling performance in real sea for 9 months, which has promising value to be used in antifouling application of marine equipment surface.

2. Results and discussion

2.1. Synthesis of PDMS-based PU with light-response coumarins releasing performance

In general, as the Scheme 2 showed, the isocyanate-terminated

PDMS (NCO-PDMS) is prepared by the reaction of hydroxypropyl terminated poly(dimethylsiloxane) (HDT-PDMS) and isophorone diisocyanate (IPDI). The polycondensation reaction with dihydroxycoumarin (CM(OH)₂), dihydroxyeugenol (EU(OH)₂) and NCO-PDMS was continued, and then linked with 1,4-butanediol (1,4-BDO) to obtain the silicone-modified PU with side chains of coumarins and eugenol. Finally, the free 4-methylcoumarin (CM) was fixed on the PU side chain by dimerization with CM(OH)₂ under 365 nm UV, released by de-dimerization under 254 nm UV. Overall, the antifouling system of modified PU coating in this work was constituted by the eugenol and coumarins fixed on the side chain of PU and the UV-responsive releasing free CM.

As showed in the Table 1, the feed mole ratio of synthetic polymer (PU-0, PU-0.5, PU-1.0, PU-1.5, PU-CM, PU-EU). Number average molecular mass (M_n), weight average molecular mass and polydispersity (Đ_M) were provided by GPC measurements in Table S1. The successful synthesis of polymer was proved by M_n of modified PU (33000 ~ 53000 g/mol), which was suitable for using as coatings.

The successful synthesis of CM(OH)₂, EU(OH)₂ and modified PU could be certified by the ¹H NMR and FTIR in Fig. S1 and Fig. S2. In addition, the FTIR of PU-1.0 after 365 nm UV treatment were in Fig. S3, the disappearance of peak of C = C in about 1600 cm⁻¹ proved the free CM has been photo-dimerized with the CM(OH)₂ on the side chain and fixed on the polymer chain.

2.2. Characterization of surface properties and transparency

According to the research of Baier, the fouling organisms attached on surface with SE of 20–30 mJ·m⁻² were easier to remove by shearing force [54]. As showed in the Fig. 1, the SE of modified PU coatings were all in the interval of 20–30 mJ·m⁻². The low SE property could be contributed to the low SE properties of siloxane chains. Moreover, the dispersion component accounts for the majority, which was due to the non-polar property of the PDMS enriched on the surface. And the SE of modified PU had no obvious relationship with the added amount of CM(OH)₂ and EU(OH)₂.

The WCA (Water Contact Angle) is an important basis for judging the hydrophilicity and hydrophobicity, reflecting the stability of the surface

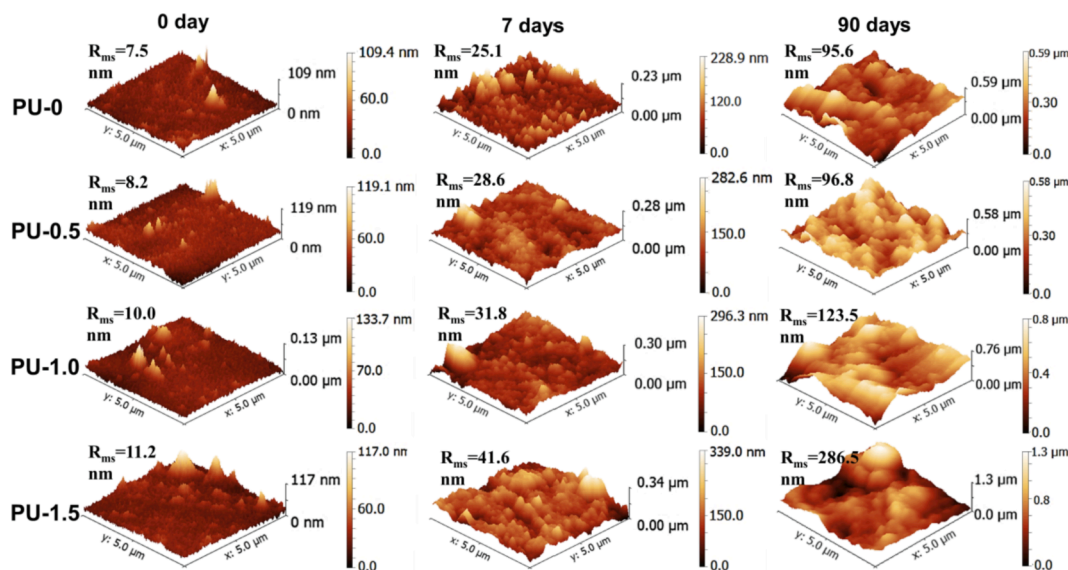
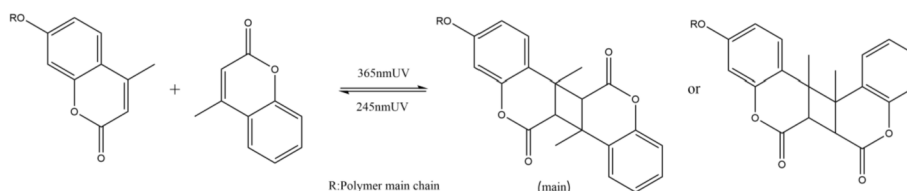


Fig. 2. AFM image of modified PU coatings immersed in sea water for 0, 7 and 90 days.



Scheme 3. UV photo-dimerization and UV de-dimerization mechanism of coumarins.

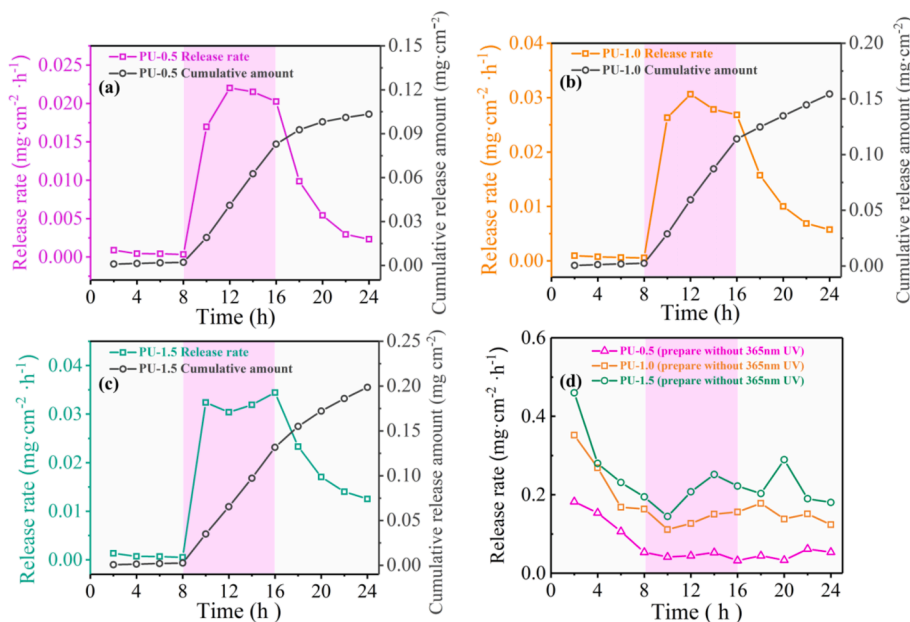


Fig. 3. 254 nm UV response CM releasing performance of (a) PU-0.5 (b) PU-1.0 (c) PU-1.5 (d) PU without 365 nm UV process.

wettability. The coatings were immersed in ASW (Artificial sea water) for 3 months in nature sunlight and the WCA trends of coatings were measured. The results show that the modified PU coatings were hydrophobic, as time went by, WCA dropped down to relatively stable values. Due to the hydrophobic benzene and lactone rings in the lipophilic coumarins and eugenol, as the content of them going by, the WCA

values increased. After immersing in ASW for 3 months, the PDMS based PU coatings still maintained hydrophobic properties ($WCA > 90^\circ$) and the WCA remained relatively stable during the process, which proved the stability of surface wettability in sea environment.

Surface roughness has a great influence on the adhesion of marine fouling organisms. Generally speaking, the rough surface provides more

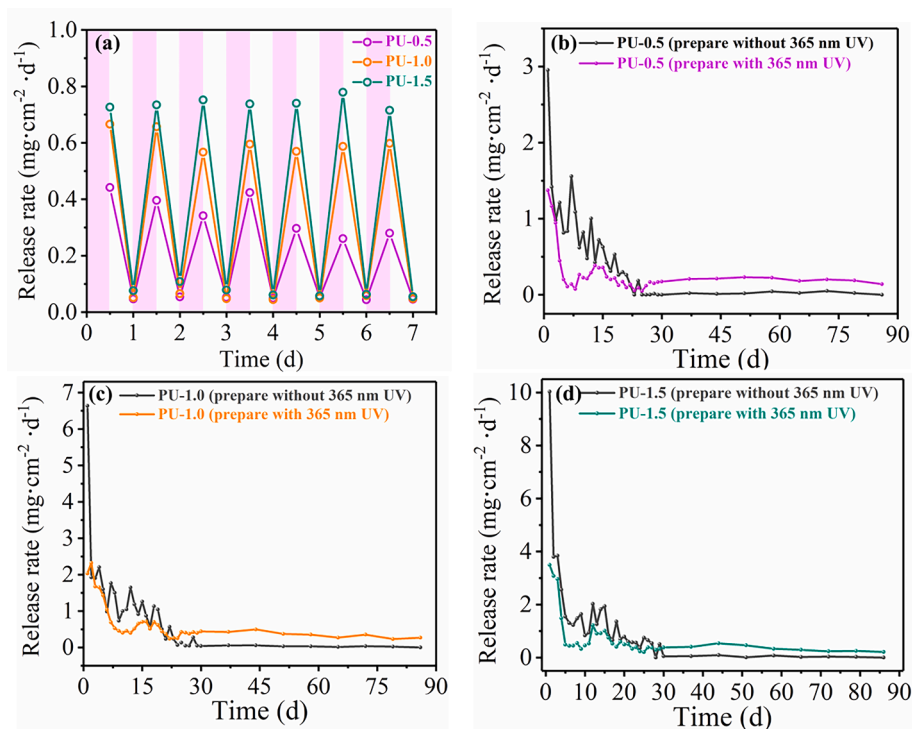


Fig. 4. (a) Cyclic CM release performance for 7 days (254 nm UV 12 h, darkness 12 h) and (b) long-term CM release performance of PU-0.5 for 3 months (day: sunlight; night: darkness) (c) long-term CM release performance of PU-1.0 for 3 months (day: sunlight; night: darkness) (d) long-term CM release performance of PU-1.5 for 3 months (day: sunlight; night: darkness).

Table 2

Element content of surface before and after immersion.

Samples	C (mol%)	Si (mol%)	O (mol%)	N (mol%)	C/Si
PU-0.5-before	51.5	24.37	23.23	0.89	2.11
PU-0.5-after	52.05	23.76	22.88	1.32	2.19
PU-1.0-before	52.43	23.92	22.87	0.78	2.19
PU-1.0-after	53.54	22.64	22.9	0.93	2.36
PU-1.5-before	53.17	23.53	22.54	0.77	2.26
PU-1.5-after	54.08	21.34	23.03	1.56	2.53

attachment sites, which facilitates the attachment and growth of fouling organisms on the surface of the equipment. For most marine fouling organisms, surfaces with low SE and low roughness have lower adhesion force with them and are easier to be removed by shear force [6]. Surface topography and roughness of modified PU coatings (just prepared, immersed in ASW for 7 days and for 90 days) were observed by AFM in Fig. 2. Because of flexibility and surface enrichment of PDMS segment,

the PDMS-based PU coating were relatively smooth with Rms of about 10 nm. And the added content of CM(OH)₂, EU(OH)₂ and CM had no apparent influence on the Rms of coatings. As the surface segment rearrangement of polar and non-polar segments, the Rms increased to 20–40 nm 7 days later and to about 100–300 nm after 90 days. Overall, prepared PU coatings could maintain low surface roughness of about 0.1–0.3 μm even after soaking in ASW 90 days, which could do contribution to reducing sailing resistance and fouling organisms.

In addition, the coating with good permeability to visible light have the potential for applications in underwater optical devices and imaging sensors. Fig. S10 showed the transmittance of PU-0.5, PU-1.0 and PU-1.5, which all exceeds 93%.

2.3. Characterization of UV responsive coumarins releasing performance

Traditional antifoulant- releasing low SE antifouling coatings generally physically mix antifouling agents into coatings, which is

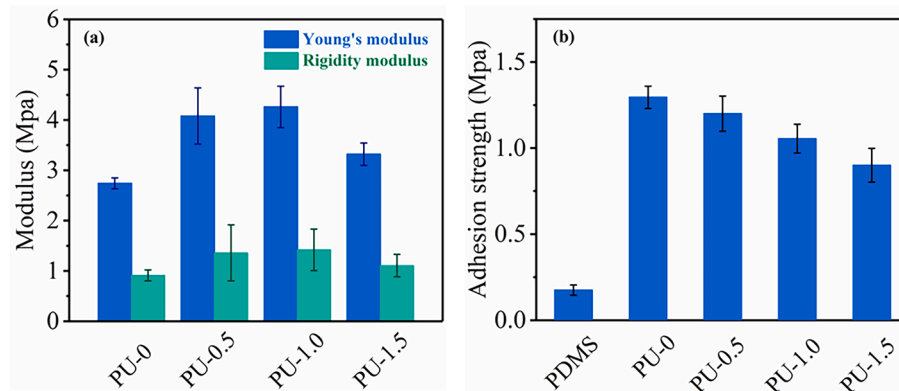


Fig. 5. (a) Modulus and (b) adhesion strength of coatings.

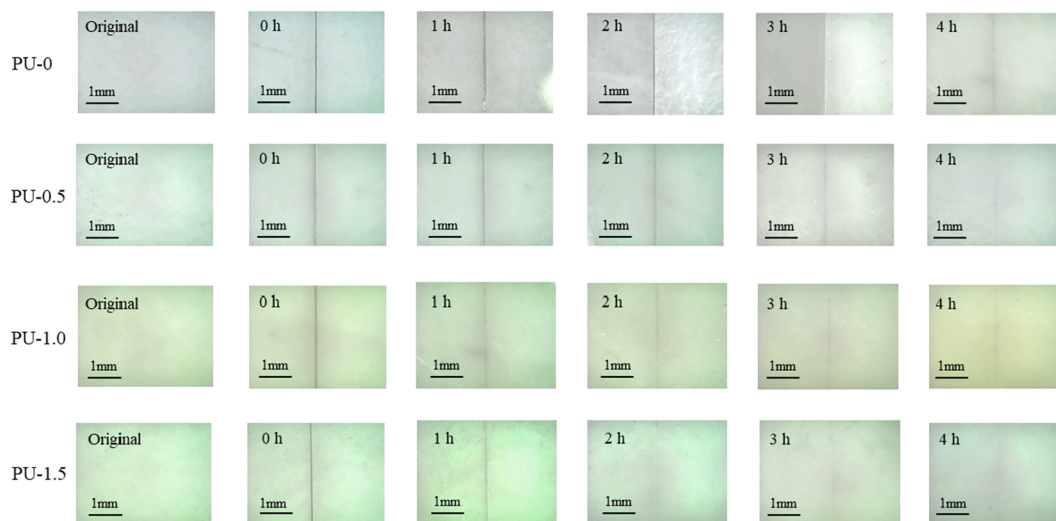


Fig. 6. Digital photos of healing process in sea water (UV 0–4 h).

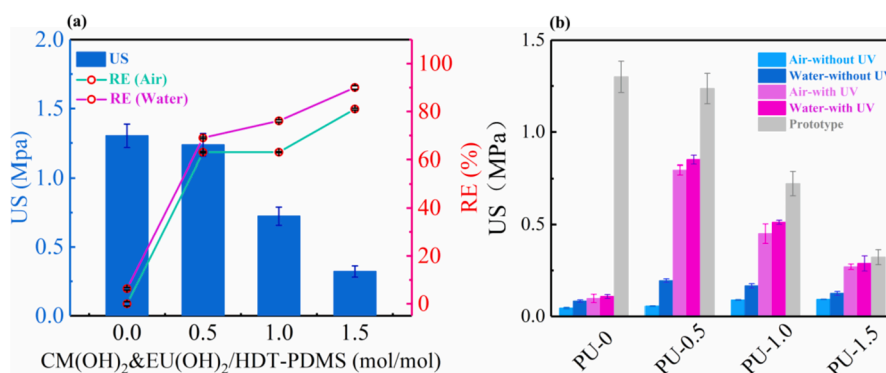


Fig. 7. (a) Repairing efficiency (RE) and ultimate strength (US) of healed PU in water (b) US of healed PU with and without 365 nm UV irradiation in water and air.

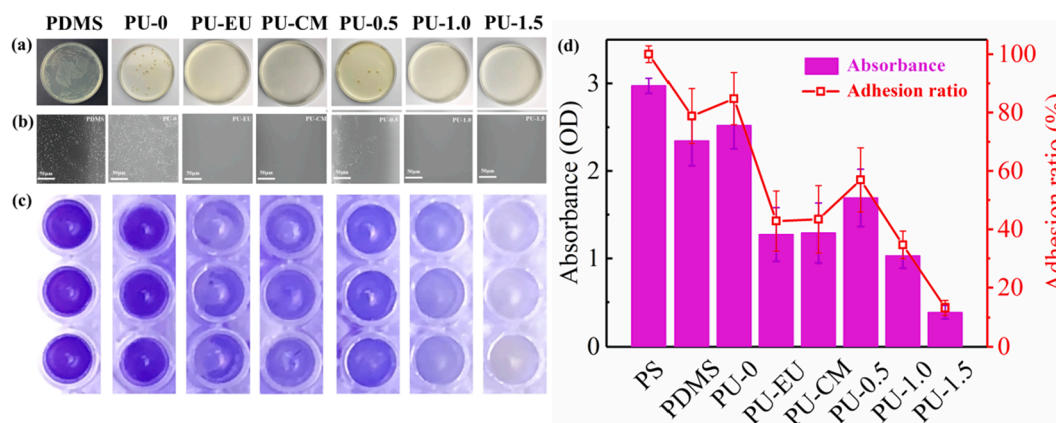


Fig. 8. Antibacterial (*E. coli*) performance of PDMS and prepared antifouling coatings with different content of antifouling agent (Illustration of PU-0 ~ 1.5, PU-EU, PU-CM was listed in Table 1). (a) Antibacterial result of LB plate method (b) antibacterial result of Electron microscope observation (c) antibacterial result of crystal violet dyeing method (d) OD value of the coating dyed by crystal violet and adhesion ratio of *E. coli* on different coatings.

difficult to achieve stable and controllable release of antifoulants. Due to changes in different anti-fouling environments and demands, smart anti-fouling coatings with controllable antifoulants releasing performance have more advantages.

As Scheme 3 showed, the PDMS-based PU with controlled release performance of coumarins antifoulant was depended on the UV photo-dimerization and UV de-dimerization behavior of coumarins. The FTIR

of PU-1.0 after 365 nm UV and then 254 nm UV treatment was in Fig. S3, the reappearance of peak of C = C in about 1600 cm⁻¹ proved that the free CM connected to the polymer chain by 365 nm UV de-dimerized.

Firstly, the UV-response CM releasing property was represented in 24 h. The coatings were immersed in ASW in darkness for 8 h, then irradiated by 254 nm UV for 8 h, following with darkness for 8 h. From Fig. 3, the CM releasing rate under 254 nm UV was significantly faster

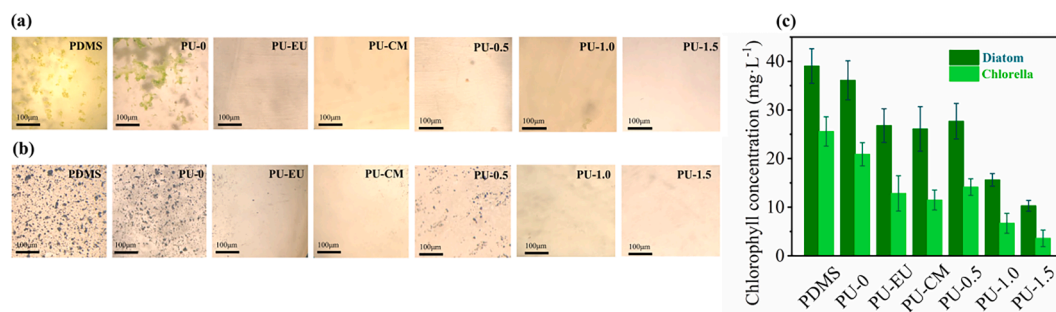


Fig. 9. Anti-algae test of prepared coatings (a) Anti-chlorella test (b) Anti-diatom test (c) quantitative test of chlorophyll.

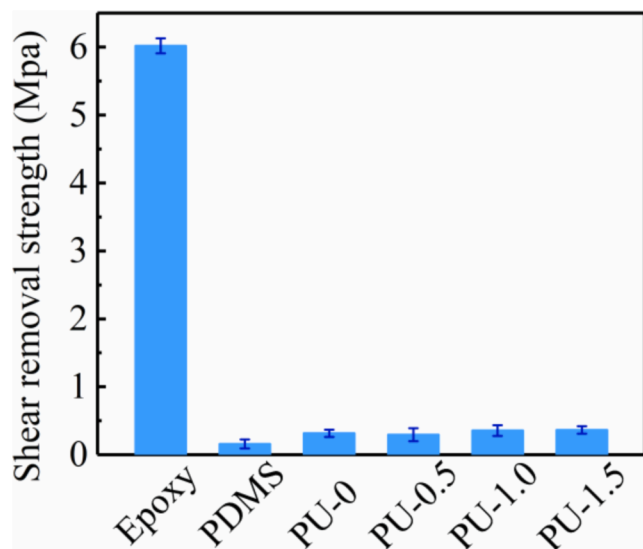


Fig. 10. Pseudobarnacle shear removal strength of epoxy, PDMS, PU-0, PU-0.5, PU-1.0 and PU-1.5 coatings.

than that in darkness, which proved that the modified PU could respond to stimulation of 254 nm UV, “unlocking” CM by de-dimerization. The process of PU polymer solution being irradiated by 365 nm UV “locked” the free CM on the polymer chain through photo-dimerization, while in the PU without 365 nm UV irradiation, the free CM was just physically compounded into the coating system and could not response to UV stimulation. So in Fig. 3 (d), the CM releasing rate was not affected by 254 nm UV in 24 h.

Then the cyclic CM release performance was tested in simulated environment of 12 h 254 nm UV irradiation and 12 h darkness per day for a week. The result in Fig. 4 (a) further showed that the release of CM could be controlled by 254 nm UV in simulated day-night cycles. As the sunlight is the most common light source in nature and includes all wavelengths, the CM releasing performance of PU synthesized with and without 365 nm UV preprocess was characterized under nature sunlight (day: sunlight; night: darkness) for 3 months. As showed in Fig. 4, without dimerization of 365 nm UV, the PU coatings which were physically compounded with CM showed serious burst-release and unstable-release behavior in the early stage of immersion. Then the CM in coatings tended to exhaust in the later period, which would weaken the antifouling property. In the contract, PU-0.5, PU-1.0 and PU-1.5 alleviated the burst release behavior, and maintained a relatively stable release rate within 3 months, proving better long-term stable releasing performance of CM.

In order to analysis the element content on coating surface and CM releasing performance before and after immersing for 3 months, XPS was tested to prove surface enrichment and release of CM on the coating

surface. In the Table 2, the element content of surface (Si, O, C, N) was measured. Because the main constituent element of CM is C and the PDMS as the main body of the coating contains Si and C, the mole ratio of C/Si could be regarded as a measure of CM migrating to surface. If more CM migrates to surface, the content of C should increase. After immersing in ASW for 3 months, the C/Si value raised up, which proofing the CM has been released from coating body to surface.

2.4. Mechanical performance

Previous studies have showed that lower modulus was beneficial to the remove of fouling organisms from the coating surface[55,56]. As show in Fig. 5, the modified PU have low Young's modulus below 4.5 Mpa. Although the addition of rigid benzene ring would increase modulus, the excess of CM(OH)₂ and EU(OH)₂ decreased the Mn and PU-1.5 had more flexible chain and lower modulus. The adhesion to epoxy substrate of PU-0.5 and PU-1.0 were both over 1 Mpa, which was tough enough to apply as marine coating [57].

2.5. UV-healing performance

Traditional PDMS-based low SE antifouling coating has shortcomings in mechanical properties, and difficult to repair after mechanical injure. As Scheme S1 showed, the UV-healing of the prepared coating was based on photo-crosslinking properties of coumarin under 365 nm UV irradiation. The UV-healing performance was tested in air and sea water environment respectively. In order to de-dimerize the coumarin on the “wound” and then re-dimerize, the coating pieces were processed with 254 nm UV irradiation of 1 h and then 365 nm UV irradiation of 1–4 h. As the digital photos in Fig. 6. (sea water), Fig. S6 (air) and Fig.S7 showed, the PU-0 had the worst healing property, while as the coumarins content increasing, the speed of crack healing also went up, which was because increasing of coumarin content promote the self-healing efficiency.

The stress-strain curves of prototype PU and healed PU were recorded in Fig. S8, which was in good agreement with the results of digital photos. Although more coumarins addition would raise repairing efficiency (RE), it also reduced ultimate strength (US) and made the material softer, which weaken mechanical behavior. Because the excessive addition of CM(OH)₂ and EU(OH)₂ reduced the chain length of the polymer, which weakened the entanglement between polymer chains.

In addition, the healing performance in sea water was slightly better than that in air, which might be attributed to hydrogen bonds in PU. In the underwater, the water environment might do contribution to the reconstruction of hydrogen bond in PU. In the underwater environment, the hydrophilic urethane bond groups were easy to rearrange and enrich in the wound section, which was beneficial to the reconstruction of the hydrogen bonds. While in the air environment, the low SE siloxane segment tended to gather in the wound section, which was not conducive to the reconstruction of hydrogen bonds. Then came to a question that between UV-dimerization and hydrogen bonds, which took the leading role in the process of healing? Fig. 7. (b) showed the US of

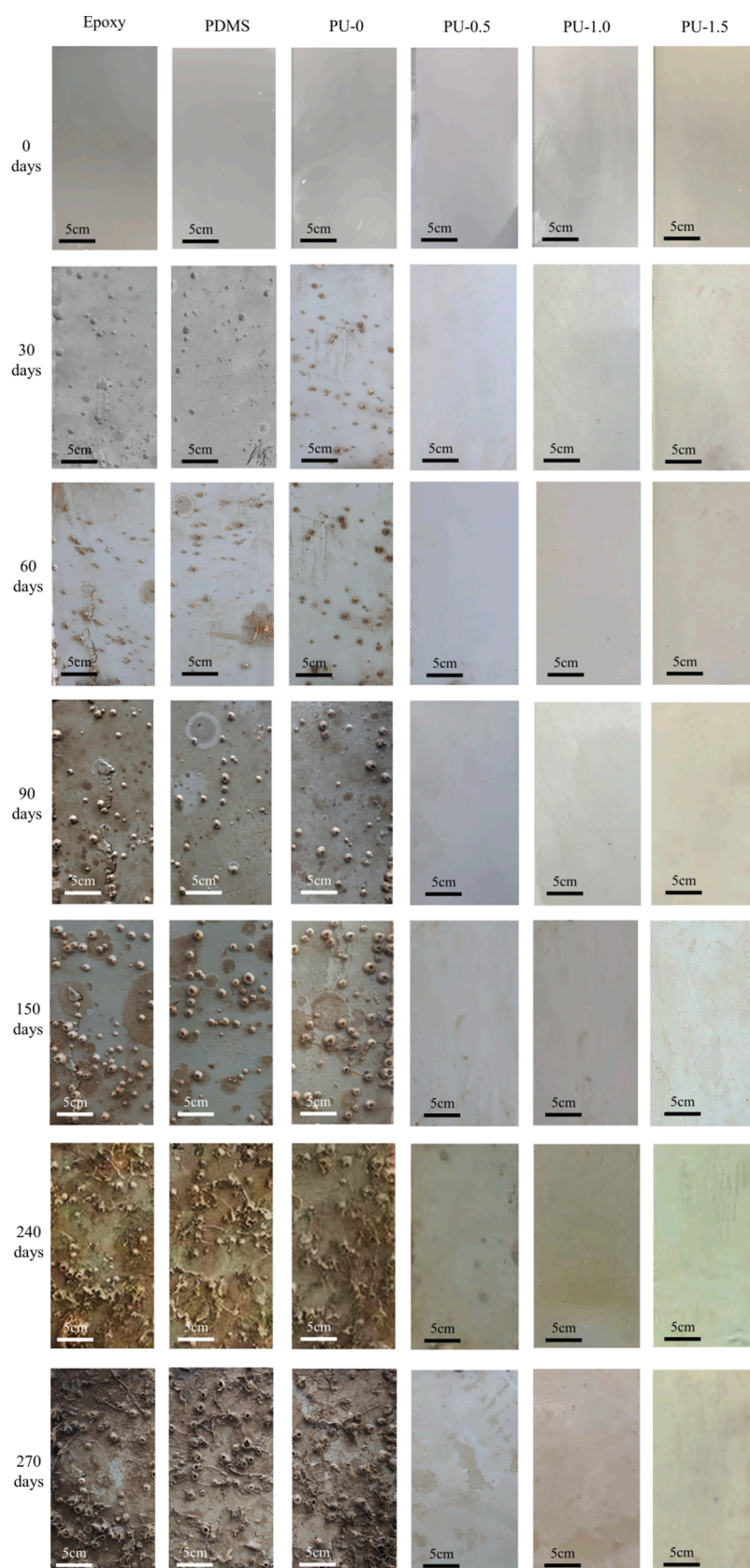


Fig. 11. Digital photos: marine field test of epoxy, PDMS, PU-0, PU-0.5, PU-1.0 and PU-1.5 coatings in Zhoushan Sea for 9 months.

healed PU which were treated with and without 365 nm UV respectively. As for the group without UV, its self-healing process only depended on hydrogen bonds. While the group with UV self-healed by both hydrogen bonds and UV-dimerization based reversible chemical bonds. The US of PU without UV process was significantly lower than that with UV process, which proving the UV-dimerization took the leading part in the healing process.

2.6. UV-antibacterial and anti-algae and pseudobarnacle removal test

Bacterial fouling is a significant part of marine fouling. As common gram-negative bacteria, *E. coli* was used to test the antibacterial performance of coatings in Fig. 8. As common gram-positive bacteria, *S. aureus* was used to test the antibacterial performance of coatings in Fig.S15. As the former research [58–60], the antibacterial mechanism of coumarins could be concluded as: (1) coumarins competitively inhibit the ATPase's activity of DNA gyrase, causing bacterial DNA to fail to form a double helix structure normally; (2) coumarins can inhibit the bacterial quorum sensing (QS) system and prevent the formation of bacterial film. QS is a mechanism for cell–cell communication and adjust gene expression in bacteria film. While the antibacterial mechanism of eugenol could be summarized as [61,62]: (1) The phenoxy anion dissociated from the phenol group forms chemical bonds with the active site of the membrane protein on cell membrane, destroying the structure and function of the bacterial membrane; (2) Eugenol can inhibit the activity of malate dehydrogenase and succinate dehydrogenase, which leads to obstruction of the redox process of the tricarboxylic acid cycle (TCA cycle) in bacterial respiration. In the results of LB plate method and electron microscope observation, the antibacterial effect increases with the addition of coumarins and eugenol. Both coumarins and eugenol had satisfactory antibacterial properties in the test of PU-CM and PU-EU. The performance of the anti-bacterial film could be quantitatively described as the OD value of the coating dyed by crystal violet, which also concided with the former results.

Anti-algae performance was tested by immersing in chlorella and diatom respectively in Fig. 9. The anti-algae mechanism of coumarins and eugenol was similar to antibacterial mechanism. And the results of Anti-algae test were also similar to antibacterial test. The result proved that both coumarins and eugenol had inhibitory effects on the growth of chlorella and diatoms. Also, with the increase of coumarins and eugenol, the attached algae decreased. The test with chlorophyll as the quantitative standard also confirmed that the antifouling coating had obvious anti-algae performance compared with PDMS and PU-0.

Pseudobarnacles of aluminum were used to evaluate the fouling release ability against barnacle in Fig. 10. The removal shear strength of modified PU was around 0.3 Mpa, which was more than PDMS but satisfactory for low SE antifouling coatings [22]. Attributed to the low SE and chemical inertness of siloxane chain, the pseudobarnacles could not adheres to the surface firmly. So pseudobarnacles responded obviously to the shear force and could be removed easily.

2.7. Marine field test

The antifouling performance of the coatings in the real epicontinental sea was examined in East China Sea near Zhoushan Islands for 9 months in Fig. 11. After immersing in seawater for 9 months, there were lots of barnacles attached on epoxy, PDMS and PU-0 coating. While the PU-0.5, PU-1.0 and PU-1.5 had nearly no fouling organisms on the surfaces, which could be owing to the antifouling effect of the inherent coumarins and eugenol in the coating and the long-term releasing free coumarin. Thus, the marine field text proved the practicality and feasibility of PDMS-based PU with controllable coumarins releasing performance in anti-fouling applications in real sea.

3. Experimental section

Details of the experimental procedures are given in the [supporting information](#).

4. Conclusion

In summary, PDMS-based PU was modified with eco-friendly coumarin compounds and eugenol to prepare the marine antifouling coating. The most optimized formulation was PU-1.0. Based on the UV dimerization and de-dimerization mechanism of coumarin, the coating could release coumarin controllably with response to UV and have the property of UV-healing. The coumarin and eugenol fixed on the polymer chain and the controllable releasing CM enhance the marine antifouling performance. Compared with traditional low SE antifouling coatings, this smart coating can release natural antifoulant controllably with response to UV and do self-healing by UV irradiation after injure, which extends the service life and enhances the application performance in real sea. The study provides a new strategy for designing “smart” antifouling coatings, which is expected to apply in marine antifouling, medical antifouling, biological antifouling and other fields.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.132471>.

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